ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. EE ENGINEERING â€" SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) Explain construction and working principle of DC generator. Derive its EMF equation. [7 Marks]
(b) What is Armature Reaction in DC machines? Explain effects of cross-magnetization and demagnetization with remedies. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) Differentiate between Lap winding and Wave winding in DC machines with simplex and multiplex diagrams. [7 Marks]
(b) Explain commutation in DC machines. How do interpoles and compensating windings improve commutation? [7 Marks]

--- UNIT II ---

Q.2 (a) Derive Torque equation of DC motor. Explain back EMF and its significance. [7 Marks]
(b) Compare speed control methods of DC shunt and series motors: Armature voltage control and Field flux control. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) Explain Swinburne's test and Hopkinson's regenerative test for finding efficiency of DC machines. [7 Marks]
(b) Why is a starter required for DC motors? Explain the operation of 3-point and 4-point starters with protective devices. [7 Marks]

--- UNIT III ---

Q.3 (a) Draw exact and approximate equivalent circuits of single-phase transformer referred to primary and secondary. [7 Marks]
(b) Explain Open-Circuit (OC) and Short-Circuit (SC) tests on single-phase transformer. How are core loss and copper loss calculated? [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) Derive condition for maximum efficiency of a transformer. Calculate all-day efficiency. [7 Marks]
(b) What is Voltage Regulation? Derive expression for voltage regulation of transformer at lagging, leading, and unity power factors. [7 Marks]

--- UNIT IV ---

Q.4 (a) What is Auto-transformer? Derive saving of copper in autotransformer compared to two-winding transformer. [7 Marks]
(b) Explain parallel operation of single-phase transformers with equal and unequal voltage ratios. State essential conditions. [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) Describe 3-phase transformer connections: Star-Star, Delta-Delta, Star-Delta, and Delta-Star with vector groupings. [7 Marks]
(b) Explain Scott Connection (T-T connection) for converting 3-phase supply to 2-phase supply with phasor diagrams. [7 Marks]

--- UNIT V ---

Q.5 (a) Explain Sumpner's (Back-to-Back) test for determining efficiency and temperature rise of two identical transformers. [7 Marks]
(b) Describe tertiary windings in 3-phase transformers and their stabilization role against unbalanced loads. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Explain tap-changing transformers: On-load and off-load tap changer operations. [7 Marks]
(b) Write short notes on: (i) Inrush current in transformers (ii) Acoustic noise and cooling methods. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****